PYTHON FOR DATA SCIENCE · LESSON 13 OF 29

Modules, Packages, Virtual Environments and Git

Organise code into modules and packages, install libraries with pip or uv inside virtual environments, pin dependencies, use the standard library, and version-control your projects with Git and GitHub.

Python class 10 | Python 3.13 · pip 25 · uv · Git 2.5x | Intermediate | 90 min

By the end of this lesson you can

- Import modules in the different ways and explain if **name** == "**main**"
- Create a package with **init**.py and a sensible project layout
- Create virtual environments and install and pin packages with pip or uv
- Use key standard-library modules (os, pathlib, datetime, random, math, statistics, collections)
- Use the everyday Git workflow: init, add, commit, branch, push, pull request

Modules

A **module** is any `.py` file. Importing it gives you its functions, classes and variables.

Python

```
import math                                  # whole module, use with prefix
from statistics import mean, median          # specific names
import datetime as dt                        # alias
from collections import Counter as C
print(math.sqrt(144), math.pi, mean([3, 5, 10]), median([3, 5, 10]), dt.date(2025, 4,
1).strftime("%d %b %Y"), C("aab"))
print(math.ceil(4.2), math.floor(4.8), math.log10(1000), round(math.exp(1), 4),
math.comb(5, 2))
```

OUTPUT

```
12.0 3.141592653589793 6 5 01 Apr 2025 Counter({'a': 2, 'b': 1})
5 4 3.0 2.7183 10
```

Style	When
`import pandas as pd`	Standard aliases (`np` , `pd` , `plt` , `sns`)
`from pathlib import Path`	A few names used often
`from module import *`	**Avoid**: hides where names come from

Writing your own module

Python

```
from pathlib import Path
Path("metrics.py").write_text('''"""Reusable business metrics."""
GST = 0.18

def margin_pct(profit, sales):
    return round(100 * profit / sales, 1) if sales else None

def with_gst(amount):
    return round(amount * (1 + GST), 2)

if __name__ == "__main__":          # runs only when executed directly, not when imported
    print("self-test:", margin_pct(25, 200))
''', encoding="utf-8")
import importlib, sys
if "" not in sys.path: sys.path.insert(0, "")        # notebooks and scripts already search
the current folder
import metrics
importlib.reload(metrics)
print(metrics.margin_pct(30, 120), metrics.with_gst(1000), metrics.__doc__)
import subprocess, sys
print(subprocess.run([sys.executable, "metrics.py"], capture_output=True,
text=True).stdout.strip())
Path("metrics.py").unlink()
```

OUTPUT

```
25.0 1180.0 Reusable business metrics.
self-test: 12.5
```

`__name__` is `"__main__"` when the file is run as a script and the module name when imported. The guard lets one file be both a library and a script.

Where does Python look?

The current folder, then the folders in `sys.path` (including `site-packages` , where pip installs libraries). `ModuleNotFoundError` usually means: wrong environment, package not installed, or a

typo. Never name your own file `pandas.py` or `random.py`: it will shadow the real library.

Packages and project layout

A **package** is a folder of modules (with an `__init__.py`). A clean data project:

Text

```
churn-project/
├── pyproject.toml          # project metadata and dependencies
├── README.md
├── .gitignore              # venv, data, secrets, __pycache__
├── data/                   # raw/ and processed/ (usually not committed)
├── notebooks/              # exploration
├── src/churn/
│   ├── __init__.py
│   ├── load.py             # read and validate data
│   ├── features.py         # feature engineering
│   └── train.py            # model training
└── tests/
    └── test_features.py
```

Import with `from churn.features import add_tenure_band`. Notebooks for exploring; `.py` modules for anything reused or scheduled.

The standard library tour

Python

```
import os, random, statistics, datetime as dt, itertools, textwrap, uuid, time
random.seed(42)
print(random.randint(1, 6), random.choice(["N", "S", "E"]), random.sample(range(100), 3),
round(random.random(), 3))
print(statistics.stdev([2, 4, 4, 4, 5, 5, 7, 9]), statistics.mode([1, 2, 2, 3]))
today = dt.date(2025, 10, 27); due = today + dt.timedelta(days=45)
print(due, due.strftime("%A"), (dt.date(2026, 1, 1) - today).days, "days to new year")
print(os.cpu_count() > 0, os.getenv("HOME") is not None, textwrap.shorten("A very long
product description here", 20), len(str(uuid.uuid4())))
```

OUTPUT

```
6 N [3, 94, 35] 0.245
2.138089935299395 2
2025-12-11 Thursday 66 days to new year
True True A very long [...] 36
```

Module	Use
`pathlib`, `os`, `shutil`, `glob`	Files and folders
`datetime`, `zoneinfo`, `time`	Dates, time zones, timing
`math`, `statistics`, `random`, `decimal`	Numbers (`decimal` for money)
`collections`, `itertools`, `functools`	Data structures and iteration helpers
`csv`, `json`, `sqlite3`, `pickle`	Data formats and storage
`re`	Regular expressions
`logging`, `argparse`, `subprocess`	Scripts and tools
`urllib`, `http`	Web (most people use `requests` / `httpx`)
`concurrent.futures`, `asyncio`	Parallel and asynchronous work

pip, virtual environments and uv

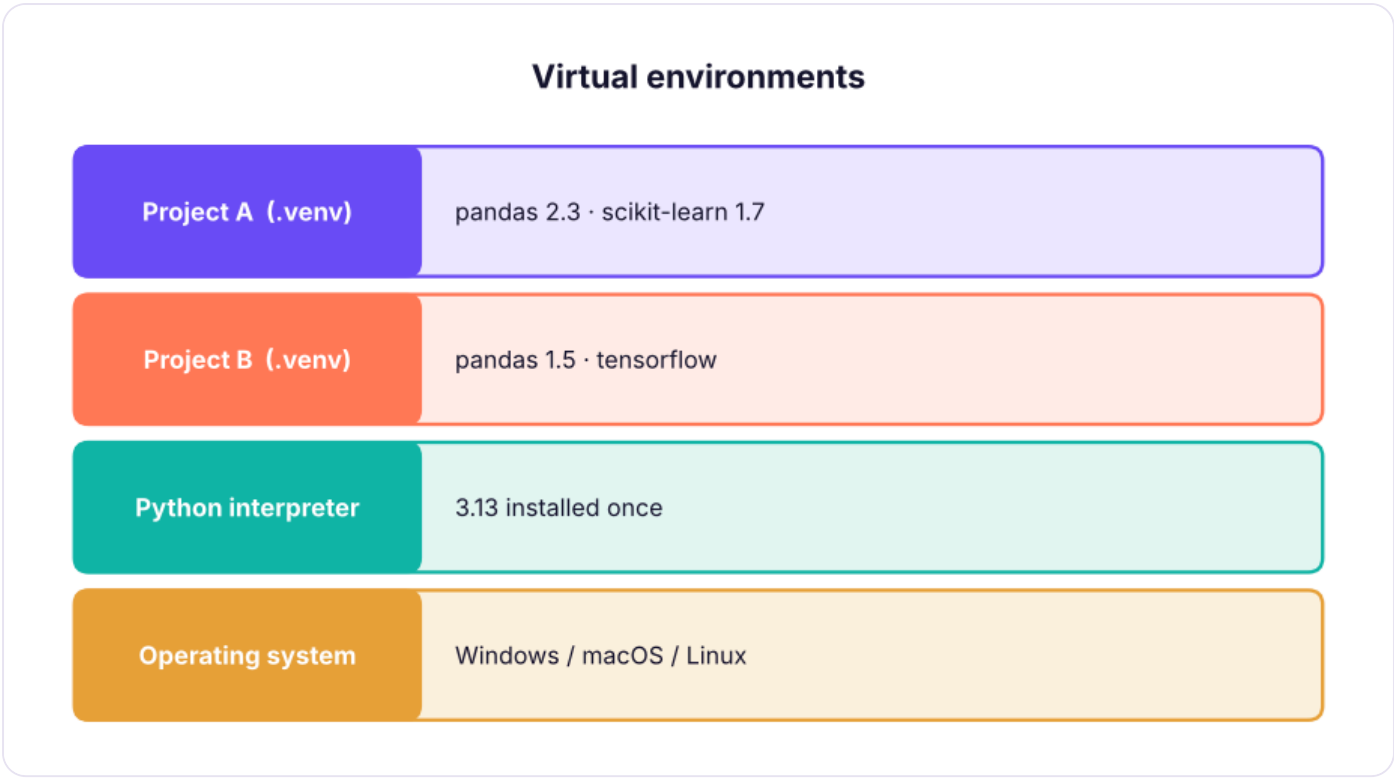
Each project should have its **own virtual environment** so its library versions do not clash with other projects.

Text

```
# Create and activate an environment (built-in venv)
python -m venv .venv
source .venv/bin/activate          # macOS/Linux
.venv\Scripts\activate             # Windows

# Install, list, pin, reproduce
pip install pandas scikit-learn matplotlib
pip list
pip freeze > requirements.txt
pip install -r requirements.txt

# uv: a much faster modern alternative (by Astral)
uv init churn-project && cd churn-project
uv add pandas scikit-learn
uv run python src/churn/train.py
```



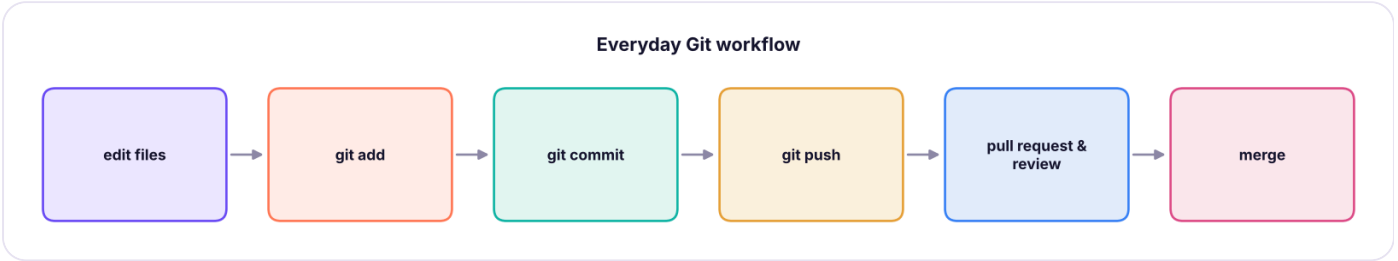
The project isolation model: each project gets its own environment and its own library versions.

TIP **Conda, Anaconda, Colab**

Anaconda/Miniconda bundle Python with data libraries and the `conda` environment manager; popular in data science courses. **Google Colab** and **Kaggle Notebooks** run Python in the browser with libraries pre-installed: ideal for practice. Pick one tool per project and stick to it.

Git and GitHub essentials

Git tracks changes to files; **GitHub** hosts repositories online for backup, collaboration and your portfolio.



The everyday Git cycle.

```text
git init                              # start tracking a folder
git status                            # what changed?
git add analysis.py                   # stage a file  (git add . = everything)
git commit -m "Add churn EDA"         # save a snapshot with a message
git log --oneline                     # history
git switch -c feature/rfm             # new branch
git push -u origin feature/rfm        # upload the branch to GitHub
git pull                              # download others' changes
git clone https://github.com/user/repo.git
git diff                              # unstaged changes
git restore file.py                   # discard local changes to a file
```

A `.gitignore` keeps junk and secrets out:

```text
.venv/
__pycache__/
.ipynb_checkpoints/
data/raw/
*.env
.DS_Store
```

WATCH OUT **Never commit secrets**

API keys, passwords and tokens belong in environment variables or a `.env` file listed in `.gitignore`. Read them with `os.getenv("OPENAI_API_KEY")`. If a key is ever pushed, revoke it immediately: deleting the commit is not enough.

Summary

KEY TAKEAWAYS

- A module is a `.py` file; a package is a folder of modules. Use standard aliases and avoid `import *`.
- `if __name__ == "__main__":` separates script code from importable code.
- One virtual environment per project; pin versions in `requirements.txt` or `pyproject.toml` (pip, uv or conda).
- The standard library covers files, dates, maths, data formats, logging and more.
- Git: add → commit → push → pull request → merge; use `.gitignore` and never commit secrets.

Quick quiz

6 questions · tap an answer to check it

1. What does `if __name__ == "__main__":` do?

A Imports main

B Runs the block only when the file is executed directly

C Defines the main function

D Speeds up the program

Answer: B. When imported, `__name__` is the module name.

2. Why use a virtual environment?

A It makes Python faster

B It isolates each project's library versions

C It is required by Git

D It encrypts the code

Answer: B. Different projects may need different versions.

3. How do you record the exact installed versions for others?

A `pip list`

B `pip freeze > requirements.txt`

C `git add .`

D `pip update`

Answer: B. Others recreate with `pip install -r requirements.txt`.

4. Why should you not name your file `pandas.py`?

A It is too long

B It shadows the real pandas library on import

C Git rejects it

D It is reserved

Answer: B. Python finds your file first.

5. Which Git command uploads your commits to GitHub?

A `git commit`

B `git push`

C `git add`

D `git log`

Answer: B. Commit saves locally; push uploads.

6. Where should API keys go?

A In the code

B In environment variables or an ignored .env file

C In the README

D In a commit message

Answer: B. Never commit secrets.

EXERCISE **Setup checklist**

(1) Create a project folder with a virtual environment and install pandas and matplotlib. (2) Write a module `cleaning.py` with two functions and import it from a notebook. (3) Create a GitHub repository, add a `.gitignore` and a README, and push. (4) Create a branch, change the README and open a pull request to yourself.

Show solution (worked answer)

Follow the commands in this lesson. Check with `python -c "import pandas; print(pandas.__version__)"` and `git log --oneline`. A good README states the question, the data source, how to run the code and the key findings with one chart.

INTERVIEW CORNER

Questions you may be asked

Q: Difference between a module, a package and a library? A module is one file, a package is a folder of modules, and a library is a collection of packages distributed together (e.g. on PyPI).

Q: How do you make a Python project reproducible? Virtual environment, pinned dependencies, a README with run instructions, fixed random seeds, version control, and data lineage.

Q: git merge vs git rebase? Merge combines histories with a merge commit; rebase replays your commits on top of another branch for a linear history. Never rebase shared branches.

Practise more online

Free, trusted places to test yourself on this topic. Links open the original site in a new tab.